STAT 527 Advanced Regression Analysis
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Simple Linear Regression with R

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Simple Linear Regression Model

The simple linear regression **model equation**:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \qquad\qquad i = 1, 2, ..., n$$

with $\mathrm{E}(\varepsilon_i) = 0$, $\mathrm{var}(\varepsilon_i) = \sigma^2$ and $\mathrm{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

The parameters are

- **intercept** β_0
- **slope** β_1, and
- **variance** of the **error** ϵ_i: σ^2.

Parameters β_0 and β_1 are also called the **coefficients**.

The errors $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are **assumed** to be

- uncorrelated ($\mathrm{cov}(\varepsilon_i, \varepsilon_j) = 0,\ i \neq j$), and
- homoscedastic ($\mathrm{var}(\varepsilon_i) = \sigma^2$ does not depend on i).

Least Squares Minimization

Least squares solution minimizes the following objective function:

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 \tag{1}$$

Differentiate the criterion (1) with respect to β_0, β_1 and set the derivatives equal to 0, we get:

$$\begin{aligned} \frac{\partial}{\partial \beta_0} &= (-2) \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial}{\partial \beta_1} &= (-2) \sum_{i=1}^{n} x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \end{aligned}$$

Solving these equations gives LSE $\hat{\beta}_0$ and $\hat{\beta}_1$.

Least Square Estimates

The minimizers $\hat{\beta}_0$ and $\hat{\beta}_1$ are called **least squares estimates** ("Hat" notation is used for estimates):

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \qquad \hat{\beta}_0 = \overline{y} - \hat{\beta}_1\overline{x}$$

Predicted Values and Residuals

Given a sample of data, how can we obtain estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ of β_0 and β_1? We will see shortly ...

For now, let $\hat{\beta}_0$ and $\hat{\beta}_1$ be estimates of β_0 and β_1, respectively.

The value

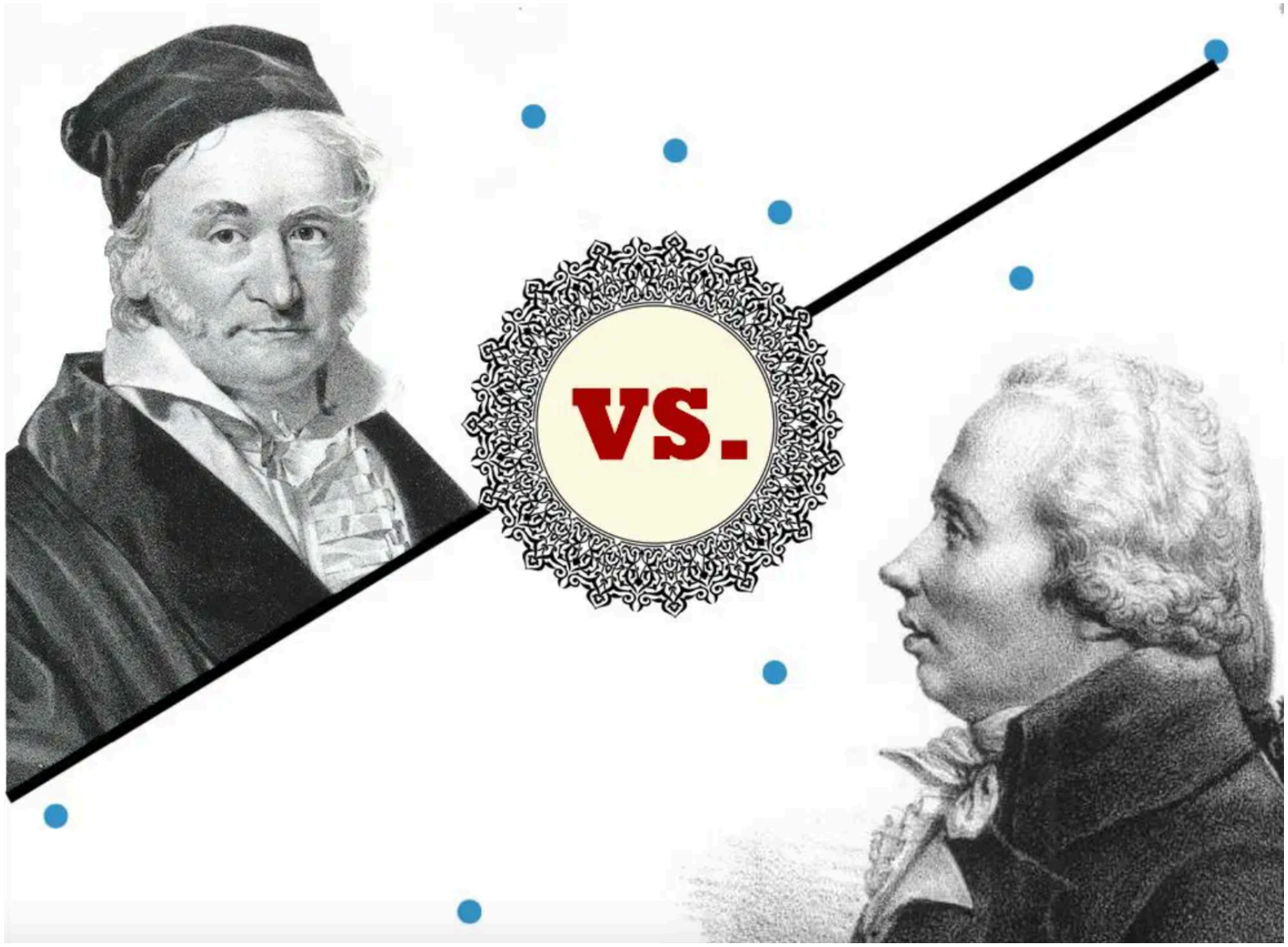
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

is called the **fitted value** (or **predicted value**) at x_i, or the **prediction** of y_i.

The ith **residual** is the difference between y_i and its prediction:

$$\hat{\varepsilon}_i = y_i - \hat{y}_i$$

Who Invented Least Squares?



source: https://priceonomics.com/the-discovery-of-statistical-regression/

Legendre's Contribution

Legendre was the first to make his discovery of the least squares method public. In his 1805 paper, “New Methods for Determination of the Orbits of Comets,” Legendre supplied the original articulation and example of the use of least squares regression. Legendre was confident that his method was a winner:

“Of all the principles which can be proposed for [making estimates from a sample], I think there is none more general, more exact, and more easy of application, than that of which we have made use… which consists of rendering the sum of the squares of the errors a minimum.”

Gauss's Contribution

In Gauss’s 1809 treatise, “Theory of the Motion of the Heavenly Bodies Moving About the Sun in Conic Sections,” he was to predict when and where the asteroid Ceres would appear in the night sky. This uses, among other things, the method of least squares.

“Our principle, which we have made use of since 1795, has lately been published by Legendre….” Gauss wrote, “where several other properties of this principle have been explained.”

to Gauss's credit, he went beyond Legendre and succeeded in connecting the method of least squares with the principles of probability and to the normal distribution.

Illustration with R

[Will illustrate fitted values and residuals ...]

R Implementation

Let's take a look at R using the melanoma example ...

```
> # Read data set from a text file in the current working directory
> Melanoma <- read.table("Skin.dat", header=TRUE)
> dimnames(Melanoma)
[[1]]
[1] "AL" "AZ" "AR" "CA" "CO" "CT" "DE" "DC" "FL" "GA" "ID" "IL" "IN" "IA" "KS"
[16] "KY" "LA" "ME" "MD" "MA" "MI" "MN" "MS" "MO" "MT" "NE" "NV" "NH" "NJ" "NM"
[31] "NY" "NC" "ND" "OH" "OK" "OR" "PA" "RI" "SC" "SD" "TN" "TX" "UT" "VT" "VA"
[46] "WA" "WV" "WI" "WY"
[[2]]
[1] "latitude" "mortality" "ocean"
> # "ocean" indicates whether or not the state borders an ocean
```

Display Data

```
> # Show the whole data set

> Melanoma
   latitude mortality ocean
AL     33.0       219     1
AZ     34.5       160     0
AR     35.0       170     0
CA     37.5       182     1
CO     39.0       149     0
CT     41.8       159     1
DE     39.0       200     1
DC     39.0       177     0
FL     28.0       197     1
GA     33.0       214     1
ID     44.5       116     0
IL     40.0       124     0
IN     40.2       128     0
IA     42.2       128     0
KS     38.5       166     0
KY     37.8       147     0
LA     31.2       190     1
ME     45.2       117     1

...
```

Linear Regression Fit

```
> # Fit simple linear regression model of mortality on latitude
> Model1 <- lm(mortality ~ latitude, data=Melanoma)
> # Print a summary of the results
> summary(Model1)
Call:
lm(formula = mortality ~ latitude, data = Melanoma)
Residuals:
Min 1Q Median 3Q Max
-38.972 -13.185 0.972 12.006 43.938
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 389.1894 23.8123 16.34 < 2e-16 ***
latitude -5.9776 0.5984 -9.99 3.31e-13 ***
---
Signif. codes: 0 `***' 0.001 `**' 0.01 `*' 0.05 `.' 0.1 ` ' 1
Residual standard error: 19.12 on 47 degrees of freedom
Multiple R-squared: 0.6798,Adjusted R-squared: 0.673
F-statistic: 99.8 on 1 and 47 DF, p-value: 3.309e-13
```

Fitted Values and Residuals

```
> # Get the fitted values and the residuals

> fitted(Model1)
      AL       AZ       AR       CA       CO       CT       DE       DC
191.9274 182.9609 179.9721 165.0280 156.0616 139.3242 156.0616 156.0616
      FL       GA       ID       IL       IN       IA       KS       KY
221.8156 191.9274 123.1846 150.0839 148.8884 136.9331 159.0504 163.2347
      LA       ME       MD       MA       MI       MN       MS       MO
202.6871 119.0002 156.0616 136.9331 129.1622 114.2181 193.1229 159.0504
      MT       NE       NV       NH       NJ       NM       NY       NC
108.2405 141.1175 156.0616 127.3689 148.8884 179.9721 132.1510 176.9833
      ND       OH       OK       OR       PA       RI       SC       SD
105.2517 148.8884 176.9833 126.1734 145.3018 139.3242 187.1453 121.3913
      TN       TX       UT       VT       VA       WA       WV       WI
173.9945 200.8938 153.0727 126.1734 165.0280 105.2517 157.2571 123.1846
      WY
132.1510

> residuals(Model1)
         AL          AZ          AR          CA          CO          CT
 27.0726285 -22.9609178  -9.9721000  16.9719894  -7.0615570  19.6758231
         DE          DC          FL          GA          ID          IL
 43.9384430  20.9384430 -24.8155502  22.0726285  -7.1845604 -26.0839213
```

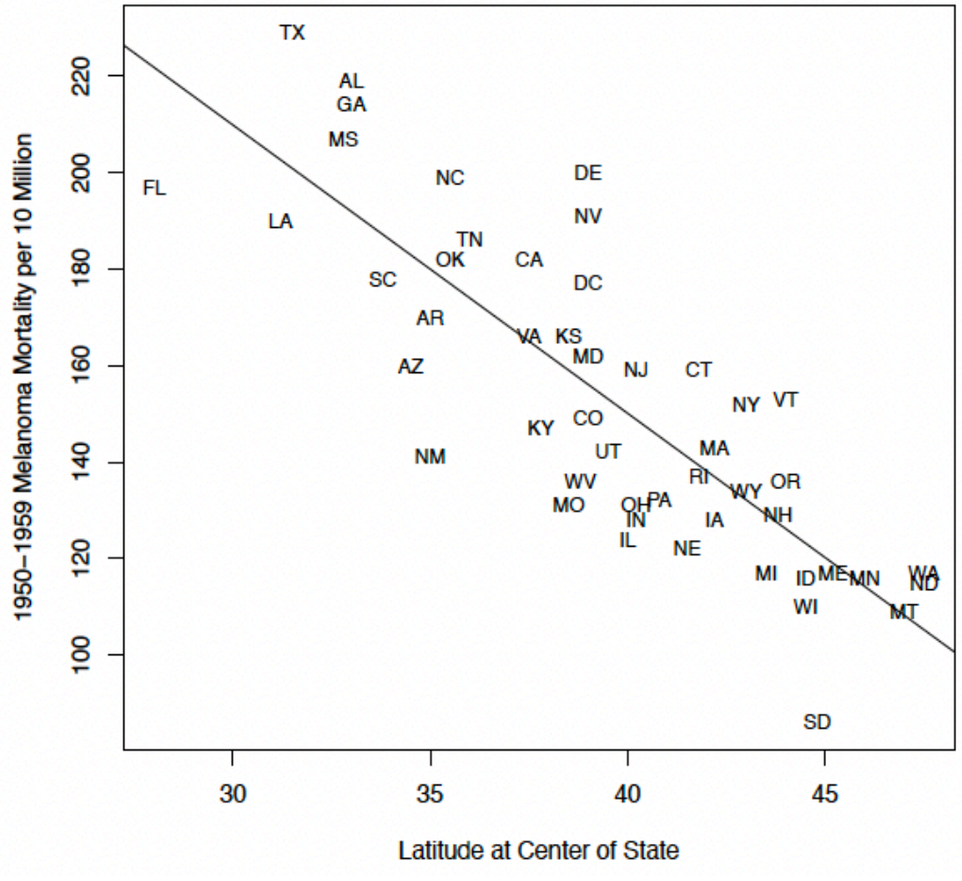
Plot

```
> plot(Melanoma$latitude, Melanoma$mortality, type="n", xlab="", ylab="")

> text(Melanoma$latitude, Melanoma$mortality, rownames(Melanoma), cex=0.8)

> title(xlab="Latitude at Center of State",
+       ylab="1950-1959 Melanoma Mortality per 10 Million")

> abline(Model1)
```



Thank you!

We gratefully acknowledge the former course instructor Naveen Narisetty and online resources for some of the images and slides presented.